2. The power P dissipated in a resistor of resistance R is calculated using the expression

$$P = \frac{V^2}{R}$$

where V is the potential difference (p.d.) across the resistor.

In an experiment to determine P, R is measured to within a percentage uncertainty of 2%.

What is the maximum percentage uncertainty in the measurement of V such that P can be determined with a maximum uncertainty of ± 5%?

A 1.5% **B** 3.0% **C** 3.5% **D** 7%

Answer: A

$$P = \frac{V^2}{R}$$

$$\Rightarrow \frac{\Delta P}{P}\frac{\Delta V}{V} = 2\frac{\Delta V}{V} + \frac{\Delta R}{R}$$

$$0.05 = 2\frac{\Delta V}{V} + 0.02$$

$$\frac{\Delta V}{V} = 0.015$$